

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_ Dry Run Evaluation

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

Register Number	192424390
Name	Venu G

COURSE OUTCOME COVERED

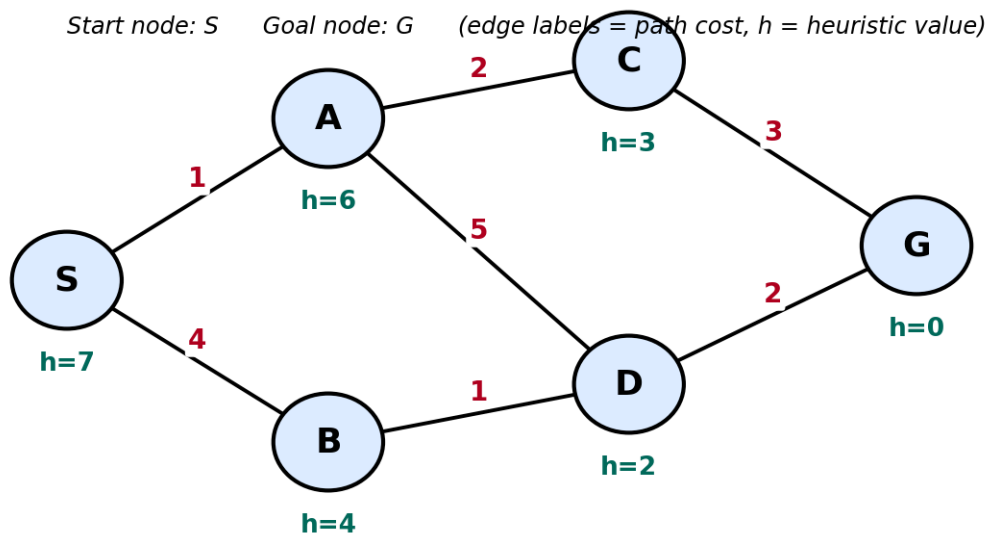
CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

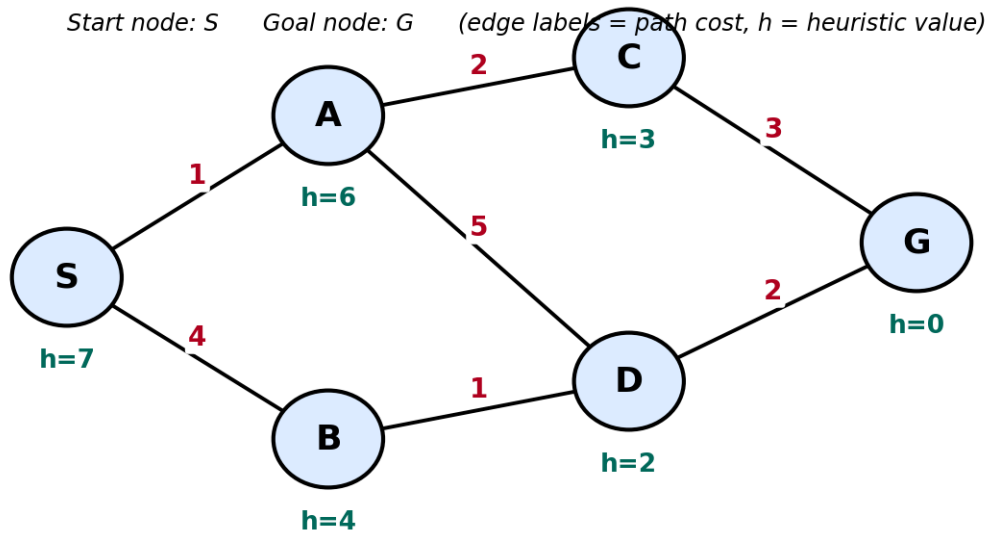
Total Marks:50

Q1. Consider a suitable state-space graph (with heuristic values $h(n)$ assigned to each node) of your choice for a given problem. Perform a dry run of Greedy Best-First Search from the start node to the goal node. Clearly show the frontier/open list, the node selected for expansion at each step, and the final path obtained.



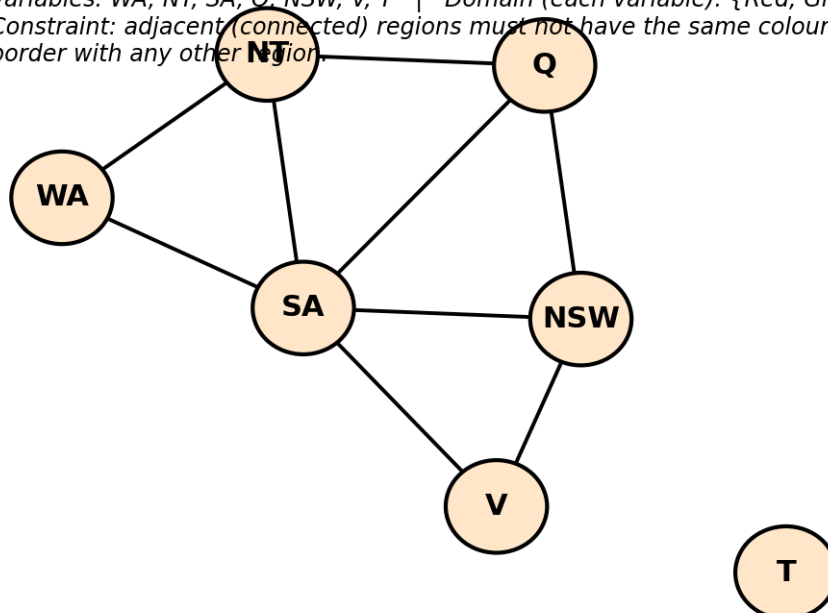
Q2. For a suitable weighted state-space graph (with edge costs and heuristic values $h(n)$) of your choice, perform a dry run of A* Search from the start node to the goal

node. Show the $g(n)$, $h(n)$ and $f(n)$ values computed at every step, the order in which nodes are expanded, and the final optimal path along with its total cost.



Q3. Formulate a given Constraint Satisfaction Problem (e.g., Map Colouring / N-Queens) by clearly stating the variables, their domains, and the constraints involved. Perform a dry run of the Backtracking Search algorithm, showing each variable assignment, every constraint check, and any backtracking steps, until a complete solution (or failure) is reached.

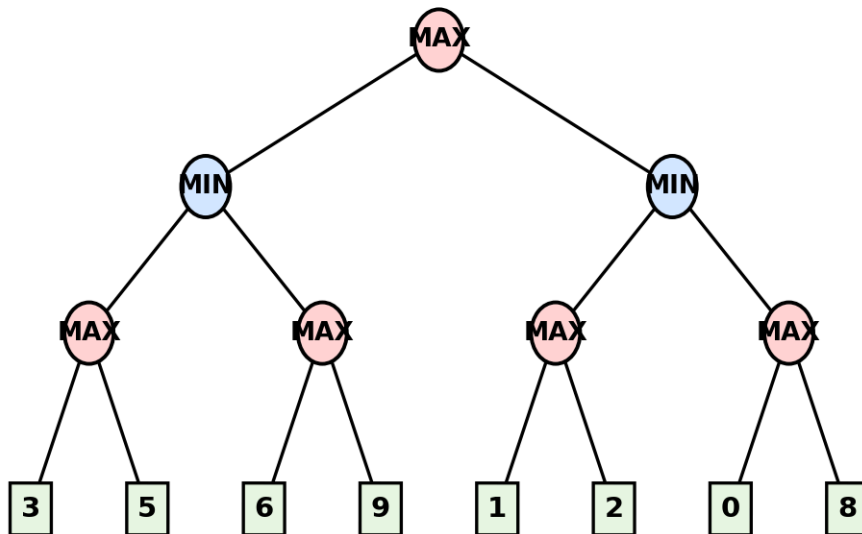
Variables: WA, NT, SA, Q, NSW, V, T | Domain (each variable): {Red, Green, Blue}
 Constraint: adjacent (connected) regions must not have the same colour. T has no shared border with any other region.



Q4. For a given two-player game tree of your choice (at least 3 levels deep), perform a dry run of the Minimax algorithm. Show the utility values at the leaf/terminal nodes

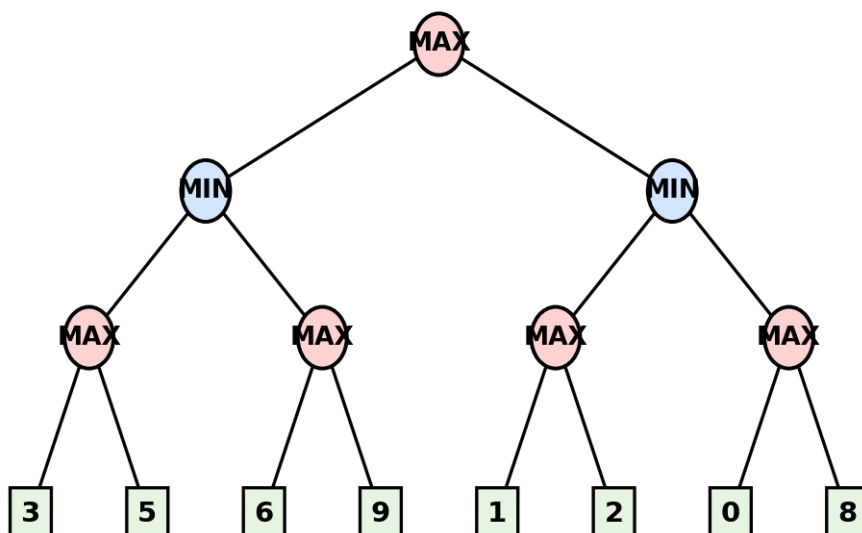
and the backed-up values computed at every internal node, and state the best move for the MAX player with justification.

Root = MAX to move. Squares at the bottom are terminal (leaf) nodes with their utility values.



Q5. Using the same (or a different) game tree from Q4, perform a dry run of the Minimax algorithm with Alpha-Beta Pruning. Show the α and β values maintained at each node during the traversal, clearly indicate which branches get pruned and why, and state the final best move obtained.

Root = MAX to move. Squares at the bottom are terminal (leaf) nodes with their utility values.



Code of Conduct:

I Venu G(192424390) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dry Run Evaluation

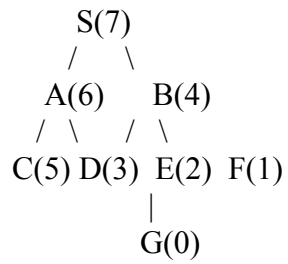
RUBRICS FOR CALCULATION

Criteria	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning	Max Marks
Problem Formulation & Setup (states/nodes, heuristic values or CSP variables, domains and constraints correctly identified)	Accurately identifies all required elements (states, heuristics/costs, or variables/domains/constraints) with clear, correct notation.	Identifies most required elements correctly, with only minor omissions or notational errors that do not affect the dry run.	Identifies some required elements; noticeable errors or missing components affect the later steps.	Little or no correct problem formulation; required elements are largely missing or incorrect.	10
Correct Application of Algorithm Procedure (expansion order, backtracking, or pruning as per the standard method)	Follows the exact algorithmic procedure at every step with no deviation from the standard method.	Follows the procedure correctly for most steps, with only one or two minor deviations that do not change the outcome.	Several steps deviate from the correct procedure, though the overall approach is recognisable.	Algorithm steps are largely incorrect, or the procedure is not followed.	10
Accuracy of Computations (g(n), h(n), f(n) values, minimax backed-up values, or α - β updates)	All computed values are accurate and consistent throughout the dry run.	Minor calculation errors are present but do not significantly affect the	Multiple calculation errors are present that affect the correctness of intermediate steps.	Calculations are mostly incorrect or missing.	10

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
		final outcome.			
Correctness of Final Outcome (final path, CSP solution, or best move obtained)	Final result is completely correct and matches the expected optimal outcome.	Final result is mostly correct, with only a minor deviation from the optimal outcome.	Final result is only partially correct.	Final result is incorrect or not obtained.	10
Clarity of Presentation & Justification (trace tables/trees/diagrams, explanation of each step)	Dry run is presented clearly using well-organised tables/trees/diagrams, with every step explained and justified.	Dry run is reasonably clear and mostly organised, with minor gaps in explanation.	Dry run presentation is unclear or poorly organised, with limited justification.	Dry run is disorganised or illegible, with little to no justification.	10
				Total	50

Q1. Greedy Best-First Search Dry Run

State Space Graph



- Start Node = S
- Goal Node = G
- Numbers in brackets are heuristic values $h(n)$.

Step 1

Open List = {S(7)}

Expand S

Children:

- A(6)
- B(4)

Open List

Node	$h(n)$
------	--------

B	4
---	---

A	6
---	---

Select B (lowest heuristic).

Step 2

Expand B

Children

- E(2)
- F(1)

Open List

Node	$h(n)$
------	--------

F	1
---	---

E 2

A 6

Select F

Step 3

Expand F

Child

- G(0)

Open List

Node h(n)

G 0

E 2

A 6

Select G

Goal Found.

Expansion Order

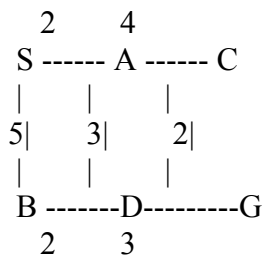
$S \rightarrow B \rightarrow F \rightarrow G$

Final Path

$S \rightarrow B \rightarrow F \rightarrow G$

Q2. A* Search Dry Run

Weighted Graph



Heuristic Values

Node	$h(n)$
------	--------

S	7
---	---

A	5
---	---

B	6
---	---

C	2
---	---

D	1
---	---

G	0
---	---

Formula

$$f(n) = g(n) + h(n)$$

Step 1

Start

Node	g	h	f
------	-----	-----	-----

S	0	7	7
---	---	---	---

Expand S

Children

Node	g	h	f
------	-----	-----	-----

A	2	5	7
---	---	---	---

B	5	6	11
---	---	---	----

Choose A

Step 2

Expand A

Children

Node	g	h	f
------	---	---	---

C	6	2	8
---	---	---	---

D	5	1	6
---	---	---	---

Open List

Node	f
------	---

D	6
---	---

C	8
---	---

B	11
---	----

Choose D

Step 3

Expand D

Child

Node	g	h	f
------	---	---	---

G	8	0	8
---	---	---	---

Open List

Node	f
------	---

C	8
---	---

G	8
---	---

B	11
---	----

Choose G

Goal Reached.

Expansion Order

$S \rightarrow A \rightarrow D \rightarrow G$

Optimal Path

$S \rightarrow A \rightarrow D \rightarrow G$

Total Cost

$2 + 3 + 3 = 8$

Q3. Constraint Satisfaction Problem (Map Colouring)

Variables

A, B, C

Domain

{Red, Green, Blue}

Constraints

$A \neq B$

$B \neq C$

$A \neq C$

Backtracking Dry Run

Step 1

Assign

A = Red

No conflict.

Step 2

Assign

B = Red

Constraint

$A = B$

Violation

Backtrack.

Assign

B = Green

Valid.

Step 3

Assign

C = Red

Constraint

A = C

Violation

Backtrack.

Assign

C = Green

Constraint

B = C

Violation

Backtrack.

Assign

C = Blue

Valid.

Final Solution

Variable	Colour
A	Red
B	Green
C	Blue

Assignment Order

A=Red

↓

B=Red ✗

↓

Backtrack

↓

B=Green ✓

↓

C=Red ✗

↓

Backtrack

↓

C=Green ✗

↓

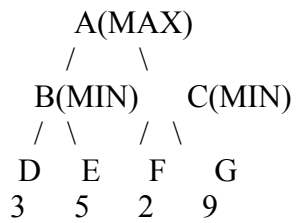
Backtrack

↓

C=Blue ✓

Q4. Minimax Algorithm Dry Run

Game Tree



Leaf utilities

D=3

E=5

F=2

G=9

Backed-up Values

Node B (MIN)

$\min(3,5)=3$

Node C (MIN)

$\min(2,9)=2$

Node A (MAX)

$\max(3,2)=3$

Final Values

A = 3

B = 3

C = 2

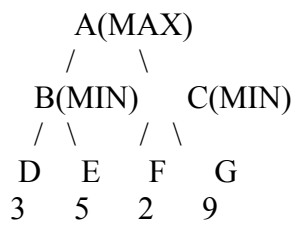
Best Move

A $\rightarrow$ B

Because MAX chooses the larger backed-up value.

Q5. Alpha-Beta Pruning Dry Run

Use the same tree.



Step 1

At A

$$\alpha = -\infty$$

$$\beta = +\infty$$

Visit B.

Step 2

At B

$$\alpha = -\infty$$

$$\beta = +\infty$$

Visit D

$$\text{Value} = 3$$

$$\beta = \min(+\infty, 3) = 3$$

Visit E

$$\text{Value} = 5$$

$$\beta = \min(3, 5) = 3$$

Return

$$B = 3$$

Step 3

At A

$$\alpha = \max(-\infty, 3) = 3$$

Visit C

$$\alpha=3$$

$$\beta=+\infty$$

Visit F

$$\text{Value}=2$$

$$\beta=\min(+\infty,2)=2$$

Now

$$\beta \leq \alpha$$

$$2 \leq 3$$

Therefore,

Prune G

(No need to evaluate G.)

Final Values

$$A = 3$$

$$B = 3$$

$$C = 2$$